

Non-Contact Blood Pressure Estimation From Radar Signals by a Stacked Deformable Convolution Network

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Abstract—This study introduces a contactless blood pressure monitoring approach that combines conventional radar signal processing with novel deep learning architectures. During the preprocessing phase, datasets suitable for synchronization are created by integrating Kalman filtering, multiscale bandpass filters, and a periodic extraction method in the time domain. These data comprise data on chest micro variations, encapsulating a complex array of physiological and biomedical information reflective of cardiac micromotions. The Radar-based Stacked Deformable convolution Network (RSD-Net) integrates channel and spatial self attention mechanisms within a deformable convolutional framework to enhance feature extraction from radar signals. The network architecture systematically employs deformable convolutions for initial deep feature extraction from individual signals. Subsequently, continuous blood pressure estimation is conducted using self attention mechanisms on feature map from single source coupled with multi-feature map channel attention. The performance of model is corroborated via the open-source dataset procured using a non-invasive 24 GHz six-port continuous wave radar system. The dataset, encompassing readings from 30 healthy individuals subjected to diverse conditions including rest, the Valsalva maneuver, apnea, and tilt-table examinations. It serves to substantiate the

validity and resilience of the proposed method in the non-contact assessment of continuous blood pressure. Evaluation metrics reveal Pearson correlation coefficients of 0.838 for systolic and 0.797 for diastolic blood pressure predictions. The Mean Error (ME) and Standard Deviation (SD) for systolic and diastolic blood pressure measurements are -0.32 ± 6.14 mmHg and -0.20 ± 5.50 mmHg, respectively. The ablation study assesses the contribution of different structural components of the RSD-Net, validating their significance in the overall of model performance.

Index Terms—Non-contact estimation, blood pressure, radar, stacked neural networks, deformable convolution.

I. INTRODUCTION

THE escalating burden of cardiovascular and cerebrovascular diseases, further compounded by the chronic impacts of COVID-19, have intensified the focus on these conditions as significant global health concerns [1]. In Intensive Care Units (ICUs), where patients are particularly vulnerable, the necessity for consistent blood pressure monitoring is definite, with traditional intermittent cuff-based methodologies falling short in providing real-time and dynamic readings [2], [3]. Cuffless blood pressure estimation via pulse wave analysis has emerged as an innovative approach, allowing for the ongoing assessment of blood pressure through correlations with Pulse Transit Time (PTT) and Pulse Arrival Time (PAT) [4], [5]. Non-contact methods still lag behind contact-based cuffless techniques in prediction accuracy due to signal instability issues [6]. While wrist sites benefit from reduced respiratory noise, they demand precise alignment with the sensor, a requirement not always feasible for all patients [7]. Chest sites, though less restrictive in patient positioning, are more susceptible to respiratory effects, necessitating sophisticated signal processing to extract reliable blood pressure data [8]. Previous studies have substantiated the correlation between chest pulse micro-vibrations and cardiac pulsations [9], [10].

In this study, we introduce an effective data preprocessing scheme for chest location physiological signal collection and a Radar-based Stacked Deformable convolution Network (RSD-Net). During the data preprocessing phase, we meticulously designed two Infinite Impulse Response (IIR) bandpass filters with different parameters for the radar phase change signals at

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the chest location, to separately capture the Chest Pulse Micro-Vibrations (CPMV) and Radar collected Heart-Beat (RHB). CPMV and RHB are derived from the same signal segment using different filtering parameters and processing methods, which will be elaborated upon in Section II-A. Concurrently, it has been substantiated that the Second derivative of the Photoplethysmogram (SPPG), often referred to as the acceleration plethysmogram, exhibits a correlation with the degree of arterial stiffness [11], [12], [13]. The CPMV similarly reflects thoracic surface movements induced by the propagation of the pulse wave. We have emulated the analytical approach applied to PPG, and computed the second differential of the CPMV signal to enrich the temporal information available for subsequent estimation algorithms [9], [10], [14]. The signal of post-calculation is designated as the Second derivative CPMV (SCPMV).

The initial module of RSD-Net incorporated a Stacked Multimodal Physiological Signal Feature Extraction Layer (SMP-Layer) for feature extraction of composite biomedical signals. Such a design enables dynamic adjustment of the convolution receptive of the kernel field and the channel weights, thereby accommodating the complexities and variabilities inherent in physiological signal processing. The backend of the model synergizes features from the SMP-Layer through the integration of a composite attention mechanism to integrate information.

This paper contributes to the field in four major ways:

- 1) Time-domain feature extraction which is predicated on the principles of hemodynamics was conducted on the CPMV signals, induced by cardiac pulsations and captured by radar, as well as on Photoplethysmogram (PPG) signals obtained from the fingertip, to ascertain their interrelatedness.
- 2) The proposed data preprocessing methods are designed to optimize the preservation of radar-collected physiological signals across a comprehensive range of feature scales, thereby enhancing the fidelity of the signal capture for biomedical analysis based on hemodynamics.
- 3) The RSD-Net employs a composite fusion framework adept at handling physiological signals, featuring an SMP-Layer and a dual-tier bottleneck cascade. This architecture enables concurrent feature extraction and leverages a parallel attention mechanism to bolster the predictive capacity of the model's performance.
- 4) The model integrates a sophisticated bi-stage bottleneck cascade, optimized for SMP-Layer outputs and enhanced with Squeeze and Excitation (SE) modules, enhancing its robustness in processing unstable radar-captured physiological signals.

The structure of this paper is as follows: Section II outlines the post-acquisition signal preprocessing techniques and dataset generation for modeling. Section III details the architecture of the RSD-Net model, featuring an SMP-Layer and an optimized dual-tier bottleneck cascade for feature extraction. Section IV assesses blood pressure measurement accuracy against British Hypertension Society (BHS) standards and evaluates model performance using extensive statistical analysis. Section V conducts ablation studies to validate the effectiveness of the modules

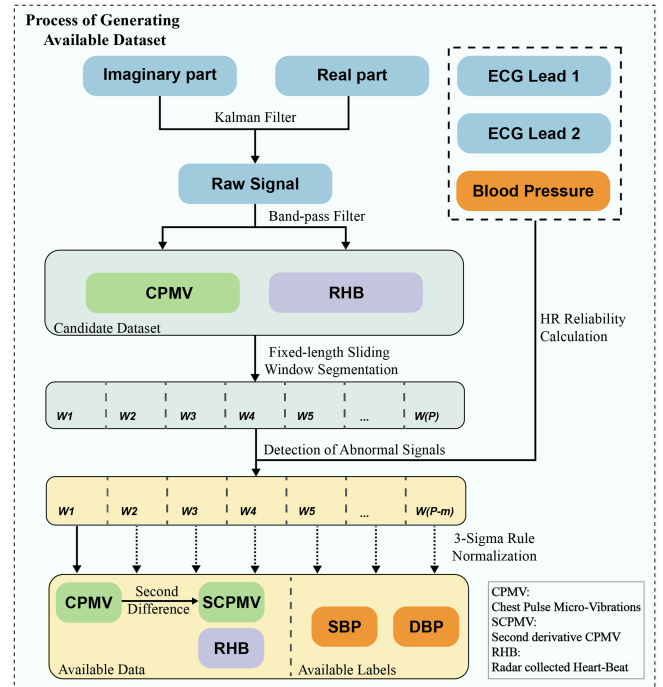


Fig. 1. Schematic of the overall process for radar-based biometric signal processing, incorporating Kalman filter demodulation and signal separation via bandpass filters with different parameters. Fixed-time window slicing ensures temporal consistency, while further preprocessing within each window including anomaly detection, resampling, application of the 3-sigma rule, and normalization yields the final usable signal dataset.

of model. Section VI concludes with a discussion of potential applications and future directions for the proposed methodology.

II. PREPROCESSING AND CORRELATION ANALYSIS OF DATA

This study draws upon the synchronous physiological signal database provided by the Institute of high-frequency technology at Hamburg University of technology [15]. The dataset consists of radar and synchronized medical data from 30 subjects, detailing subtle phase fluctuations in the chest captured by radar, Electrocardiogram (ECG), and blood pressure. The radar and ECG signals were sampled at 2000 Hz, and blood pressure at 100 Hz. We aligned the data using upsampling interpolation for the blood pressure measurements. The radar operates at a continuous wave frequency of 24 GHz. This comprehensive dataset covers various conditions such as resting, Valsalva maneuver, apnea, tilt-up, and tilt-down tests [16]. On the other hand, these actions-generated instabilities pose significant challenges to the stability and reliability of the model under various signal scenarios.

To elucidate the signal processing procedure, Fig. 1 delineates the comprehensive workflow from the real and imaginary components of the radar signal to the generation of a usable signal dataset. Following demodulation via a Kalman filter, the signal is segregated into physiological signals using bandpass filters of varying scales. Subsequent fixed-time window slicing ensures temporal consistency, within which further preprocessing of the data is conducted. This includes anomaly detection, CPMV

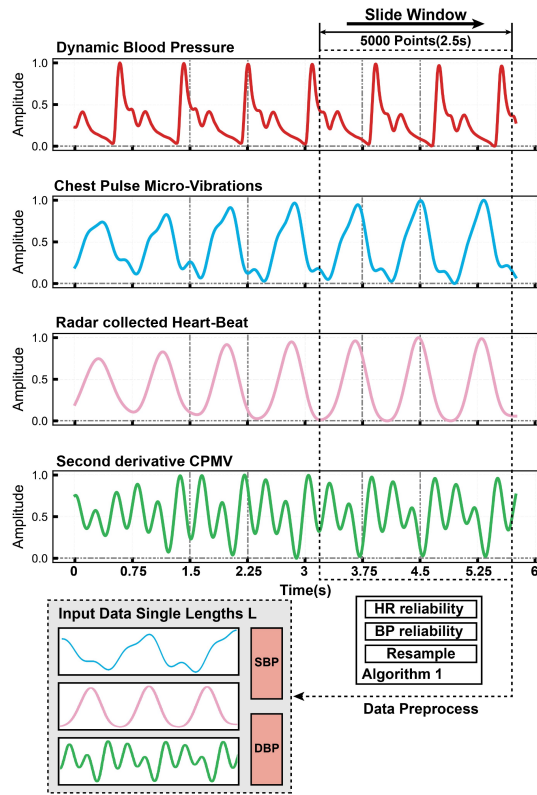


Fig. 2. Intuitive data processing alignment examples of time-domain waveforms for CPMV, RHB, SCPMV and corresponding label within a fixed window length after radar signal demodulation.

period extraction, resampling, 3-sigma rule application, and normalization steps to produce the final usable signal set. A specific example of the post-demodulation signal processing is presented in Fig. 2. The specifics of the algorithmic processing will be elaborated in the following sections of this Section.

A. Physiological Signal Preprocessing

Radar signals directed towards the chest area can capture a variety of physiological signals, each corresponding to a different frequency component range. Considering the complex attributes of the acquired signals, the application of bandpass filtering across various frequency ranges facilitates the extraction of representations pertinent to the original radar signals on different temporal scales, encompassing both the CPMV and RHB signals.

In the phase demodulation approach, Kalman filtering was applied and initialized by the state transition and the observation matrix. For the CPMV signal, referencing the frequency range of the PPG signal, a 12th-order IIR bandpass filter with a frequency band of 0.8 Hz to 5 Hz was employed [17]. The RHB signal constitutes a narrower frequency band input feature complement to the CPMV signal, offering a more purified cardiac rhythm signal. Consequently, a fourth-order IIR bandpass filter with a frequency range of 0.9 Hz to 3 Hz was deployed. Considering baseline drift in long-term signal acquisition and

Algorithm 1 CPMV Signal and Blood Pressure Validation via Multilead ECG Comparisons and Range Checks

- 1: Compute HR from different ECG leads using the inverse of the R-peak intervals:
- 2: $\text{ecg1_HR} \leftarrow 60/\text{R-peak interval of ECG Lead 1}$
- 3: $\text{ecg2_HR} \leftarrow 60/\text{R-peak interval of ECG Lead 2}$
- 4: **if** $|\text{ecg1_HR} - \text{ecg2_HR}| > 20$ **then**
- 5: Choose ECG HR closer to radar-measured HR as reference:
- 6: $[_, \text{idx}] \leftarrow \min(|[\text{ecg1_HR}, \text{ecg2_HR}] - \text{radar_signal_HR}|)$
- 7: $\text{Ref_HR} \leftarrow [\text{ecg1_HR}, \text{ecg2_HR}][\text{idx}]$
- 8: **else**
- 9: Compute mean ECG HR as reference:
- 10: $\text{Ref_HR} \leftarrow \text{mean}([\text{ecg1_HR}, \text{ecg2_HR}])$
- 11: **end if**
- 12: Compute relative error:
- 13: $\text{relative_error} \leftarrow |\text{radar_signal_HR} - \text{Ref_HR}|/\text{Ref_HR}$
- 14: **if** $\text{relative_error} \leq 0.1$
- 15: and $85 \leq \text{SBP} \leq 150$ and $55 \leq \text{DBP} \leq 100$ **then**
- 16: Consider CPMV signal and corresponding BP measurements reliable.
- 17: Resample CPMV signal to 512 Hz for further processing.
- 18: **end if**

physical displacement-induced interference, employing fixed-length segmentation with a sliding window on filtered signals is essential for reliable phase change detection. The data within the window serves as a candidate set for anomaly detection, as described in Algorithm 1. Primarily, the reliability of Heart Rate (HR) and blood pressure labels is calculated through the presence of dual-lead synchronous ECG signals, enabling us to calculate the heart rate using the synchronous ECG signals from both pathways. The reliability of the radar-acquired signals is assessed by comparing the heart rate calculated from the ECG with that calculated from the CPMV.

To ensure the reliability of heart rate values extracted from radar and ECG signals, we propose a dual-lead heart rate computation algorithm designed to optimally capture the valid current heart rate. This algorithm employs two ECG leads to corroborate the heart rate derived from radar signals for reliability. By comparing heart rates calculated from distinct ECG leads, potential anomalies within CPMV signals collected by radar can be appropriately identified and addressed. The specific procedure is outlined in steps 1 to 11 of Algorithm 1. Initially, heart rates are calculated using signals from two different ECG leads, and the difference between the two values is determined. If the difference exceeds 20 beats, it is presumed that one of the rates is unreliable, and the heart rate from the radar-collected CPMV signal is then compared, selecting the rate with the smaller discrepancy as the current reference heart rate. Conversely, if the difference is less than 20, both are considered reliable, and their average is taken as the current reference heart rate. Subsequently, this reference

heart rate is compared with the rate obtained from the radar CPMV for error analysis. Should the discrepancy be less than 10%, which is 0.1 in proportional terms, the segment of the signal is incorporated into the usable signal dataset, corresponding to steps 12 to 17 in Algorithm 1.

In accordance with the American Heart Association (AHA) and American Society of Hypertension (ASH), the normal blood pressure range for healthy adults is defined as follows: Systolic pressure is considered normal when it ranges from 90 to 130 mmHg, and diastolic pressure is normal when it is between 60 and 90 mmHg. Values exceeding these ranges may be classified as prehypertension or hypertension, whereas values below are recognized as indicative of hypotension [18], [19], [20].

Fig. 2 depicts the aligned dynamic blood pressure signals obtained via synchronous acquisition interpolation and fixed time windows, as well as the CPMV, RHB, and SCPMV signals acquired using the different bandpass filter parameters mentioned above. Utilizing the data preprocessing steps described in Fig. 1, we achieved data alignment and clear representation in the time domain.

In the process of data processing, we implemented dynamic range screening for blood pressure measurements to address inaccuracies inherent in the data collection process. As all participants in the dataset were declared healthy, both excessively high and low blood pressure readings are considered to be the result of labeling measurement errors [15]. Notwithstanding this, the possibility of undetected prehypertensive conditions within the group was considered. To accommodate this, we expanded the conventional parameters for normal blood pressure, thereby improving the sensitivity of model to potential hypertensive cases without significantly compromising specificity. Consequently, the threshold for systolic blood pressure (SBP) was established between 85 and 150 mmHg and diastolic blood pressure (DBP) was set to range from 55 to 100 mmHg. This strategic calibration ensures a robust dataset, enhancing the reliability of our analysis in detecting true blood pressure deviations.

The dataset that passed the aforementioned screening is employed as the usable dataset for the peak-cutting of CPMV. In PPG signal, the peak with the highest amplitude within a cycle is a prominent feature. This corresponds to the amplitude at the end time of the feature F1 of PPG in Fig. 3(a). The signal segmentation approach employed in this study utilizes these highest peaks for feature identification, defining the period of an input CPMV by the duration of three consecutive peaks of maximum amplitude. The remaining signals, including SCPMV and RHB, directly utilize the complete slide windowed time-domain input signals. The final signal from the aforementioned three pathways is subjected to 512 points resampling, 0-1 normalization, and storage into the final usable dataset following the 3-sigma rule as the input data.

B. Correlation Analysis Between the CPMV and PPG

To rigorously establish the correlation between the novel CPMV signal and the concurrently acquired PPG signal at a computational level of hemodynamic assessment, we executed

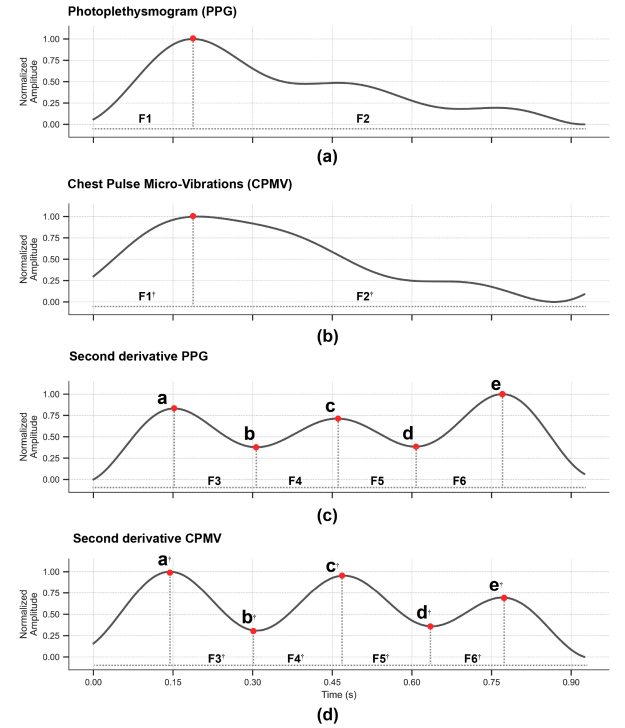


Fig. 3. Temporal waveform feature values of PPG and SPPG and schematic representation of feature matching positions for CPMV and SCPMV signals acquired by radar under similar temporal waveform.

a methodical experiment. Recognizing that the CPMV signal encapsulates phase variations in the thoracic region resulting from pulsatile wave fluctuations, we replicated the collection method of the open-source dataset which was used in this paper to conduct a non-invasive thoracic phase mapping experiment utilizing millimeter-wave radar technology. The millimeter wave radar (IWR1443, Texas Instruments (TI), USA) was the instrument of choice for signal capture, while synchronous PPG data was harvested employing the acquisition system in concert with the PPG sensor (HK-2013/2GH-L and HKG-07 C, HUAKE, CHINA). To conclusively demonstrate the correlation, a breath-holding maneuver was employed during the acquisition of the CPMV signal with a duration of 7.87 seconds approximately equivalent to eight cardiac cycles. All procured signals underwent a low-pass filtering process at a threshold of 3.5 Hz, utilizing an 8th-order filter to ensure signal fidelity. The subsequent analytical phase encompassed the computation of the Pearson correlation coefficient R across the raw waveform data, their second derivative counterparts, and respective characteristic indices.

The R calculation formula is as follows:

$$R = \frac{\sum_{i=1}^N (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_{i=1}^N (y_i - \bar{y})^2} \sqrt{\sum_{i=1}^N (\hat{y}_i - \bar{\hat{y}})^2}} \quad (1)$$

The $\hat{y}_i$ and y_i respectively signify the feature values for the CPMV and PPG signals at the i temporal waveform cycle. $\bar{\hat{y}}$ and $\bar{y}$ are the mean feature values for the CPMV and PPG datasets

TABLE I
CALCULATION OF R BETWEEN PPG TIME-DOMAIN FEATURES AND
CORRESPONDING CPMV TIME-DOMAIN FEATURES

Feature Comparison	Pearson Correlation Coefficient
F1 and F1 [†]	0.9681
F2 and F2 [†]	0.3303
F3 and F3 [†]	-0.2943
F4 and F4 [†]	0.9147
F5 and F5 [†]	0.8464
F6 and F6 [†]	0.8677
Sum F3-F6 and Sum F3 [†] -F6 [†]	0.9737
b/a and b [†] /a [†]	0.6956
d/a and d [†] /a [†]	0.8210

over all cycles, while N represents the total count of temporal waveform cycles under consideration.

Following the methodologies in references [13], [21], [22], [23], [24], we have selected PPG temporal features that are associated with hemodynamics and the degree of arterial stiffness. Corresponding temporal features for CPMV were identified to find their respective feature values.

Fig. 3 displays time-domain waveforms and second derivatives of PPG signals from a fingertip in (a) and (c). The CPMV signals and their second derivatives from radar in (b) and (d) of Fig. 3. The extracted metrics encompass temporal intervals from the onset to the initial systolic apex of the PPG signal, delineated as F1, and the terminal time point F2 as shown in Fig. 3(a). Intervals between the successive peaks and troughs of the SPPG, denoted as F3 through F6, are detailed in Fig. 3(c). The amplitude ratios of the SPPG, specifically b/a and d/a, have been previously demonstrated to be indicative of arterial stiffness. [12], [13], [25]. The temporal features of the PPG waveform were scrutinized, with equivalent landmarks on the radar-derived CPMV signal identified and uniquely labeled with the dagger symbol ([†]). Fig. 3(b)(d) exhibits these annotated CPMV waveform characteristics. A comprehensive correlation analysis of the described features, contrasting CPMV and PPG signals across the data acquisition span, is presented in Table I.

The detailed correlation calculations are presented in Table I, where Sum(F3-F6) signifies the summation of feature values from F3 to F6, previously study identified as being associated with PTT [26]. It is observable that, aside from a weak correlation exhibited between F2 and F2[†] as well as F3 and F3[†] due to the narrow range of feature value variations, the remaining feature values demonstrate robust correlation coefficients. This substantiates the association of CPMV signals, acquired by radar, with PPG signals obtained from the fingertip in computing feature values pertinent to hemodynamics.

III. RSD-NET MODEL STRUCTURE

A. Overview of the Model Structure

In the input layer, the input data consists of three types of physiological signals related to hemodynamics: CPMV, SCPMV, and RHB. In the input layer structure, we use deformable convolution kernels of different sizes designed for specific signals to extract

features on different scales. Each module contains a deformable convolution layer. The deformable convolution layer is equipped with a set of learnable parameters that dictate the spatial offsets for adaptive convolutional processing. These parameters were refined by the gradient descent algorithms, wherein the partial derivatives of a mean squared error (MSE) loss function with respect to the iterative updates of offsets. The offset convolution fundamentally serves to deduce and encode spatial transformations within the input feature space. These transformations intrinsic to the geometry of data are captured as offset values that facilitate an adaptive receptive field. Leveraging these learned offsets, the deformable convolution layer implements a convolutional mechanism that dynamically conforms to the spatial input, effectively altering the convolution kernel's spatial sampling to deal with the signal's structural intricacies. Detailed parameters are delineated in Table II, where the specific model architecture parameters are respectively listed. The lower right corner provides a clear annotation to illustrate the particular significance of each parameter.

B. Stacked Multi-Modal Physiological Signal Feature Extraction Layer

Owing to the distinct waveform characteristic information in the time domain, the CPMV, SCPMV, and RHB signals each present unique profiles. In this work, we anchor convolutional kernel size design on the CPMV signal, in conjunction with the down sampling and period extraction operations performed during data preprocessing. We propose an SMP-Layer designed to concurrently consider hemodynamic information from multiple input physiological signals. This layer employs a set of data-driven and learnable deformable convolutional kernels to adaptively extract features, thereby better accommodating the inherent variability present in human physiological signals.

Prior research has proven that PTT and PAT parameters are strongly correlated with the hemodynamic information associated with blood pressure [27], [28], [29]. Specifically, the PAT encompasses the addition information of PTT and the pre-ejection period of the heart, i.e., the additional delay between the left depolarization of the ventricle and the beginning of mechanical ventricular ejection. This duration approximates half of the entire PPG cycle [30], [31]. As the data preprocessing has downsampled the data to 512 dimensions with three CPMV and the highest peak cycles, the deformable convolution kernel of the CPMV pathway is set to 128 to accommodate corresponding waveform characteristics. As the SCPMV represents the second-order derivative of the CPMV signal and since it reflects the temporal acceleration of CPMV, the distribution of its temporal feature values is inherently more complex compared to the original waveform [32], [33]. The convolutional receptive field is set to half of the CPMV convolution kernel with 64. The RHB component features a narrower passband frequency compared to the CPMV component. We aim for it to capture a broader range of cardiac motion information, thus a longer convolutional kernel has been employed for feature extraction. With the resampling rate in mind, the convolution kernel size is set to 256, aiming to maximize the extraction of the compound

TABLE II
DETAILED PARAMETERS OF RSD-NET FOR NON-CONTACT BLOOD PRESSURE ESTIMATION

Module Name	Layer Name	Output Size	Parameter
Stacked Multi-modal Physiological Signal Feature Extraction Layer	Conv _{Deform1}	(32, 512)	(256, 1, 1, 32)
	Conv _{Deform2}	(32, 512)	(128, 1, 1, 32)
	Conv _{Deform3}	(32, 512)	(64, 1, 1, 32)
	Max Pool	(96,256)	(3, 2,...)
Bottleneck ₁	Conv1 ₁	(192,256)	(1, 1, 96, 48)
	Conv1 ₂		(3, 1, 48, 48)
	Conv1 ₃		(1, 1, 48, 192)
	Shortcut	(192, 8)	(1, 1, 96, 192)
	Multi-head Attention		(192, 8)
	Group Normal		(1, 192)
Bi-stage Enhanced Bottleneck Cascade	Conv2 ₁	(384,128)	(1, 2, 192,96)
	Conv2 ₂		(3, 1, 96, 96)
	Conv2 ₃		(1, 1, 96, 384)
	Shortcut	(384, 8)	(1, 1, 192, 384)
	Multi-head Attention		(384, 8)
	Group Normal		(1, 384)

Conv,Pool:(kernel_size,stride,channel_in,channel_out)
Multi-head Attention:(embed_dim,num_heads)
GroupNormal:(num_groups, num_channels)

relationship features. Finally, the feature maps obtained from the stacked structure are concatenated along the channel dimension and subjected to max pooling to reduce the computational load and extract salient features, which are then forwarded to subsequent layers of the model for further processing. The offsets and masks are initially generated by a convolutional layer as

$$O = \text{conv}_{\text{offset}}(X) = X * W_o + b_o \quad (2)$$

where X denotes the input feature map and W_o, b_o are the weight and bias of $\text{conv}_{\text{offset}}$, respectively. Subsequently, O is divided into the offset and mask as

$$\text{offset, pre}_{\text{mask}} = \text{split}(O, \text{split}_{\text{size}}) \quad (3)$$

The pre_{mask} is then constrained between (0, 1) using the Sigmoid function to get the mask:

$$\text{mask} = \text{Sigmoid}(\text{pre}_{\text{mask}}) \quad (4)$$

Following the mask generation, the deformable convolution is performed. It starts by applying bilinear interpolation between the input feature map X and the offset to yield the offset feature map:

$$X_o = \text{bilinear}_{\text{interpolate}}(X, \text{offset}) \quad (5)$$

The offset feature map is then element-wise multiplied with the mask and subsequently convolved with the convolutional kernel to get:

$$Y = \text{conv}_{\text{deform}}(X_o \odot \text{mask}) = (X_o \odot \text{mask}) * W_d + b_d \quad (6)$$

where $\odot$ signifies element-wise multiplication, W_d and b_d are the weight and bias of $\text{conv}_{\text{offset}}$, respectively.

Finally, the output undergoes the SE module which begins with a global average pooling:

$$Z = \text{avgpool}_{\text{global}}(Y) \quad (7)$$

Then, the scale factor is computed via two Fully Connected layers (FC) followed by ReLU and Sigmoid activation functions.

At last, the scale factor is element-wise multiplied by the original feature map to obtain the final output:

$$Y = \text{Sigmoid}(\text{FC}(\text{ReLU}(\text{FC}(Z)))) \odot Y \quad (8)$$

C. Bi-Stage Enhanced Bottleneck Cascade

The module comprises two sets of three enhanced bottleneck convolution structures each, incorporating channel and self-attention mechanisms to adapt to the variability in physiological signals. The architecture integrates spatial, multi-head, and SE modules for improved performance. The operation is as follows:

$$\mathbf{y}' = s(\mathbf{x}) + \text{shortcut}(\mathbf{x}) \quad (9)$$

Herein, $s(\cdot)$ signifies the SE module. The $\text{shortcut}(\cdot)$ denotes the shortcut operation that serves to recover and match feature dimensions, facilitating subsequent operations. Subsequently, we transform the dimensions of $\mathbf{y}'$ into a form suitable for the input of the multi-head attention module for the next signal processing step. Each head computes the independent Q, K, and V, where the Q_m, K_m , and V_m of the m -th head are generated by applying distinct linear transformations to the input feature maps, as denoted by:

$$Q_m = W_{Q_m} X, \quad K_m = W_{K_m} X, \quad V_m = W_{V_m} X \quad (10)$$

where X is the input feature map, and W_{Q_m}, W_{K_m} , and W_{V_m} are learnable weight matrices corresponding to query, key, and value in the m -th head. Once these matrices are generated, the attention scores can be calculated by taking the dot product of the query and key matrices, followed by a softmax operation for normalization:

$$S_{mj} = \text{softmax} \left(\frac{Q_m K_j^T}{\sqrt{d_k}} \right) \quad (11)$$

These scores are then used to weigh the value vectors, producing the output for each head:

$$A_m = S_{mj} V_j \quad (12)$$

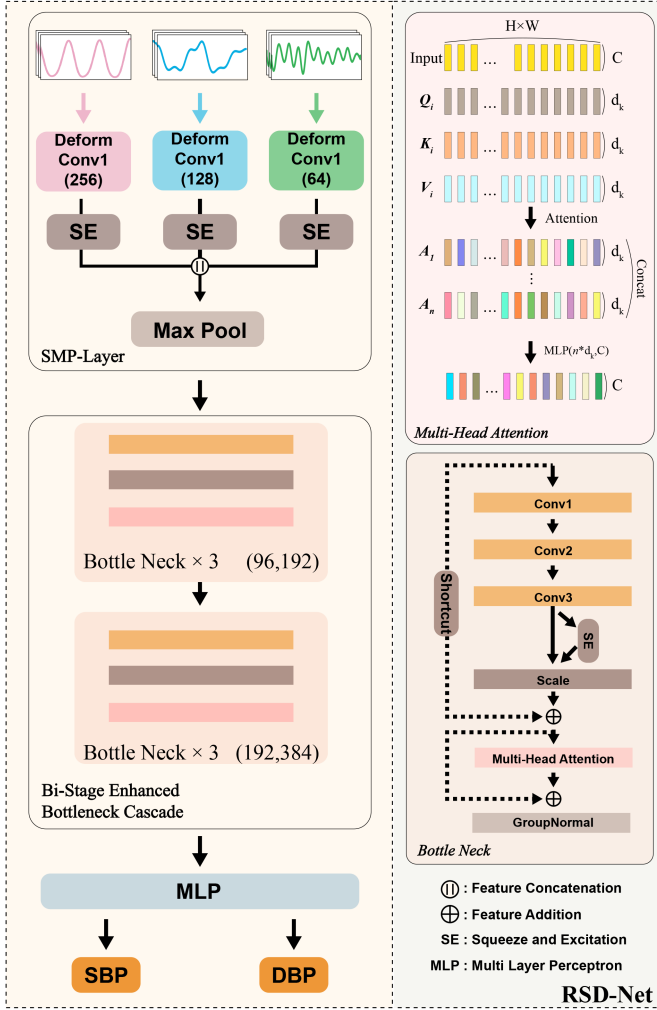


Fig. 4. Structural overview and key components of RSD-Net for blood pressure estimation.

Here, A_m is the output of the m -th head, and $\sqrt{d_k}$ is the scaling factor, where d_k is the dimension of the key. Subsequently, the outputs of all the heads $A_1, A_2, \dots, A_M$ are concatenated and linearly transformed through a fully connected layer to form a multi-head attention output of dimensions $H \times W \times C$ (Spatial $\times$ Channel Dimension). In the concluding stage of the bi-stage enhanced bottleneck cascade, we adopt a group normalization operation. The comprehensive model structure is illustrated in Fig. 4. The bottleneck structure is characterized by a pair of numbers, (96, 192) and (196, 392), which respectively represent the number of input and output channels.

IV. RESULTS

The experimentation of the proposed model was conducted on the open-source 24 G continuous-wave radar signal dataset, based on a six-port structure [15]. This open-source dataset incorporates multiple categories of synchronized human physiological signals, including the phase variation signals of the test subject's chest and corresponding dynamic blood pressure values. Five different types of signal collection were performed

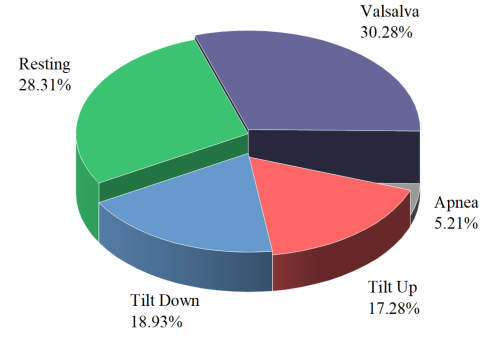


Fig. 5. Schematic diagram of data distribution proportion in different action states during acquisition.

under multiple scenarios and breathing patterns while the test subject lay on an inclined platform connected to a reference monitoring device.

A. Dataset

Specifically, the dataset used in this study includes 30 subjects, with 14 males and 16 females. The subjects' ages ranged around 30.7 ± 9.9 years, height approximately 175.7 ± 10.5 cm, and weight around 72.2 ± 14.0 kg. All subjects were declared healthy, and the data was collected by physicians from the university hospital of Erlangen following a prescribed protocol. Different scenarios included the rest, Valsalva, apnea, and tilt table tests. The distribution of data across various data collection scenarios is presented in Fig. 5. The radar measurement system captured the phase changes in the chest of the subject and simultaneously obtained the in-phase (I) and quadrature (Q) components. These were digitized concurrently at a sampling rate of 2000 Hz using a 24-bit ADC (ADS1298, TI, USA). Following the sampling, the data generated by the ADC was stored in the Infineon XMC4500 microcontroller. Kalman filtering was used to phase demodulation. Accounting for the considerable variations in hemodynamics between individuals, identical blood pressure designations can correspond to distinct physiological signal features. Hence, a subset of data was also extracted from each evaluated sample for the purpose of model training [34], [35], [36], [37]. Due to the fewer subjects number, the final distribution between the training and test sets constituted a ratio of 60% and 40% respectively. The demodulated phase signal, after undergoing preprocessing as described in Section II-A, resulted in the generation of a usable signal set.

B. Performance Evaluation of the Model Under the Blood Pressure Standards

According to the requirements of the BHS, the performance of the proposed model was assessed by the Mean Error (ME), Mean Absolute Error (MAE), and Root Mean Square Error (RMSE). The definitions of ME, MAE, and RMSE are as follows:

ME, MAE and RMSE are calculated as:

$$ME = \frac{1}{K} \sum_{k=1}^K (y_k - \hat{y}_k) \quad (13)$$

TABLE III
PERFORMANCE OF THE RSD-NET ACROSS VARYING INPUT SIGNAL LENGTHS

Number of input signal $L = 4$				Number of input signal $L = 6$				Number of input signal $L = 8$			
SBP		DBP		SBP		DBP		SBP		DBP	
MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
5.71	7.35	4.95	6.30	5.17	6.53	4.53	5.75	4.86	6.14	4.42	5.50

MAE:Mean Absolute Error; RMSE:Root Mean Square Error

TABLE IV
PERFORMANCE OF RSD-NET WITHIN THE FRAMEWORK OF THE BHS STANDARDS

Cumulative Frequency of Error (%)	<5 mmHg	<10 mmHg	<15 mmHg
Grade A	60	85	95
Grade B	50	75	90
Grade C	40	65	85
SBP prediction ($L = 4$)	53.83	82.92	95.22
DBP prediction ($L = 4$)	58.74	89.62	97.81
SBP prediction ($L = 6$)	55.74	87.30	97.80
DBP prediction ($L = 6$)	61.09	92.21	98.77
SBP prediction ($L = 8$)	62.84	88.25	98.91
DBP prediction ($L = 8$)	61.20	93.17	99.45

$$MAE = \frac{1}{K} \sum_{k=1}^K |y_k - \hat{y}_k| \quad (14)$$

$$RMSE = \sqrt{\frac{1}{K} \sum_{k=1}^K (y_k - \hat{y}_k)^2} \quad (15)$$

Where y_k represents the true value, $\hat{y}_k$ represents the predicted value, and K is the total number of samples. As shown in Table III, We have evaluated the performance of the model under the aforementioned statistical parameters with different input data lengths L , to further substantiate the robust performance of the model under varying input lengths. When the input data is set to $L=8$, the proposed model delivers its optimal performance.

Following the BHS standard, which classifies measurement accuracy based on the cumulative frequency of errors, a systematic statistical analysis was performed on data of varying input lengths. As demonstrated in Table IV, the model achieved an 'A' grade on the BHS scale when processing the longest input signals.

C. Analysis of Individual Blood Pressure Tracking and Statistical Results

The evaluation of blood pressure in different states for an individual sample is significant [38]. To ascertain the single-sample tracking performance of the proposed methodology, we performed blood pressure tracking tests on an individual subject. The states of activity included resting, Valsalva, apnea, tilt up, and tilt down. To obtain outputs with heightened resolution and better tracking performance, we adopted a moderate input signal cycle length, calculating blood pressure every six input cycles which means $L=6$ which corresponds to 15 seconds. It also entails a greater susceptibility to fluctuation in the estimated values. As illustrated in Fig. 6, the proposed RSD-Net is capable of tracking the gross trend of single-sample blood pressure

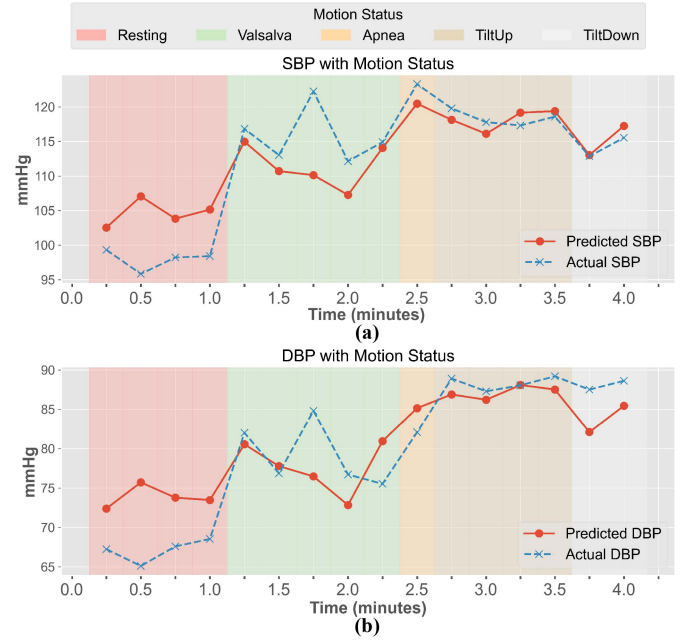


Fig. 6. SBP (a) and DBP (b) variations in individual subjects across different motion states and corresponding tracking by the RSD-Net algorithm.

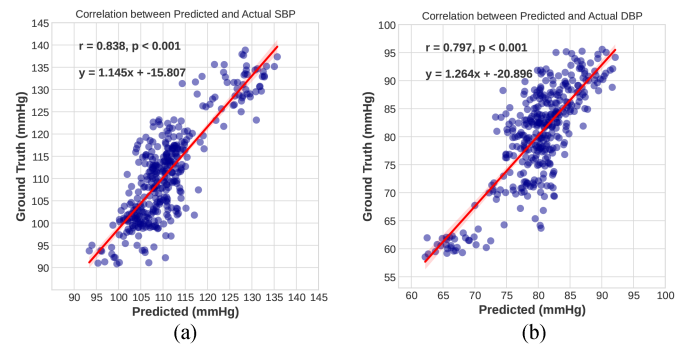


Fig. 7. Correlation of predicted and ground truth values for SBP (a) and DBP (b).

variations. However, it fails to provide accurate blood pressure estimates during abrupt changes, as observed at 0.50 and 1.75 minutes. This limitation may stem from a deficiency in the training dataset, specifically a lack of representative feature samples or labels corresponding to such rapid transitions.

Fig. 7 provides an extensive visualization of the performance of the RSD-Net model in predicting SBP and DBP, where each data point represents a comparison of predicted values and their corresponding ground truths. The fit line discloses a significant

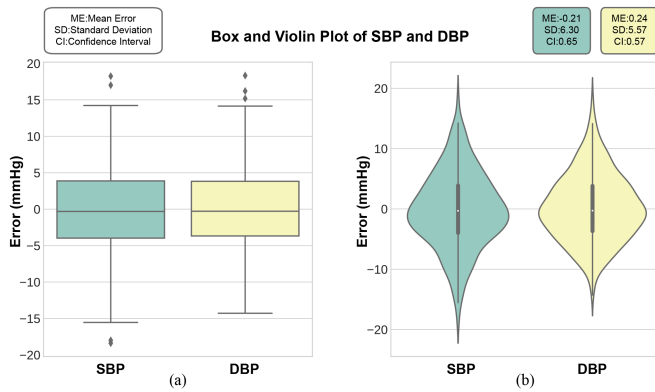


Fig. 8. Box (a) and violin (b) plot analysis of prediction errors SBP and DBP.

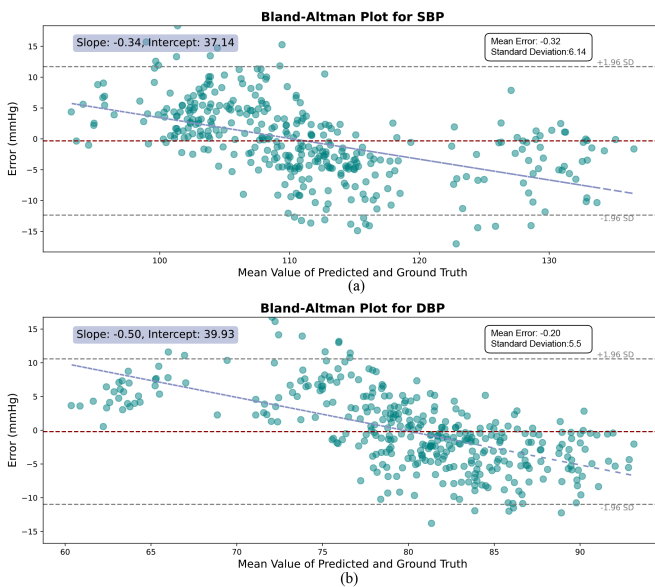


Fig. 9. Bland-Altman plots visualizing prediction errors for SBP (a) and DBP (b).

linear correlation between predicted and actual values. The confidence interval displayed denotes the range of uncertainty in prediction. The correlation coefficient annotated in the top left, alongside its p-value, confirms a statistically significant association between predicted and actual values. Furthermore, the slope and intercept of the regression line present a precise quantitative relation of this association.

Fig. 8 predominantly represents a statistical comparison of the prediction errors for SBP and DBP generated by the RSD-Net model, articulated through boxplots and violin plots. The visualizations reveal the distribution of errors for both SBP and DBP. The mean, SD, and Confidence Intervals (CI) of the error for each pressure prediction are annotated in the upper right corner, providing a comprehensive understanding of the model's performance.

Fig. 9 illustrates a Bland-Altman plots, commonly utilized for examining the agreement between two measurement methods. Each scatter point represents a single sample, with the abscissa

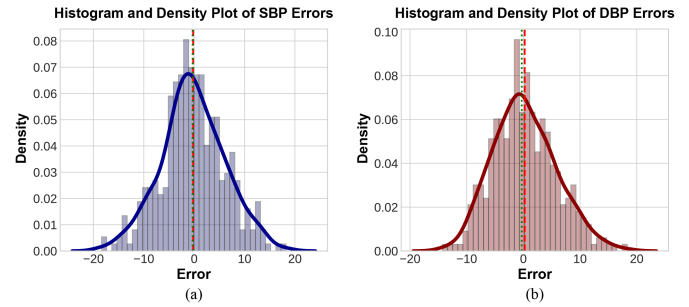


Fig. 10. Systematic error distribution analysis for SBP (a) and DBP (b) measurements.

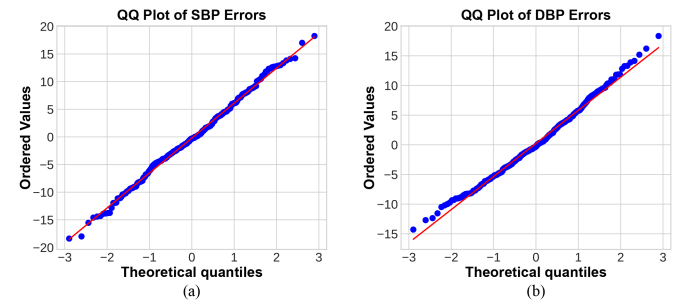


Fig. 11. Normality assessment of SBP (a) and DBP (b) measurement errors via Quantile-Quantile plots.

corresponding to the mean of true and predicted values, and the ordinate to the difference between predicted and true values. The red dashed line signifies the ME, while the grey dashed lines represent the 95% confidence interval for the error, that is, the mean value ± 1.96 SD. The text annotations within Fig. 9 provide the statistical parameters of the error, including the ME and SD. Based on the error fitting parameters in the top left corner, SBP and DBP exhibit a relatively consistent bias. It can be observed that there is an overestimation for lower blood pressure value readings and an underestimation for higher blood pressure value readings, which may stem from the conservativeness introduced by the MSE loss function.

The Fig. 10 left (a) and right (b) panels respectively display the histogram and density plot for the errors in SBP and DBP measurements. Within these plots, the light blue and light red regions indicate the distribution of errors for SBP and DBP, respectively. The dark blue and dark red lines represent density plots, depicting the probability density of the errors. The red dashed line within the graphs symbolizes the mean of the errors, and the green dotted line represents the median, providing an intuitive understanding of the central tendency of the error distribution. Observation of the aforementioned images reveals that the output errors of the model generally conform to a normal distribution.

The Quantile-Quantile (QQ) plots in Fig. 11 depict the distribution of prediction errors for SBP (a) and DBP (b) in a visually intuitive manner. By comparing the data points to the theoretical line, which represents the expected quantiles assuming a normal distribution, the skewness of the error distribution offers

TABLE V
STATISTICAL METRICS-BASED PERFORMANCE COMPARISON OF MODELS

Year	Method	Position	Subject	SBP (mmHg)						DBP (mmHg)					
				ME	RMSE	SD	MAE	MRE (%)	R (unitless)	ME	RMSE	SD	MAE	MRE (%)	R (unitless)
2019	Linear Regression [39]	wrist & neck	6	-5.3	-	5.1	-	5.10	-0.79	-1.1	-	7.09	-	8.40	-0.82
2020	Reflective PTT [40]	wrist	1	0.55	-	5.45	-	-	-	-2.26	-	3.93	-	-	-
2021	Bagging SVM [41]	chest	8	-	-	-	-	8.30	-	-	-	-	-	8.04	-
2022	Multilayer Perceptron [42]	wrist & chest	27	-	-	12.13	9.51	-	0.78	-	-	8.12	6.69	-	0.77
2022	Linear Regression [43]	chest	4	0.07	6.05	-	-	-	-	-0.07	4.8	-	-	-	-
2022	Random Forest [44]	chest	3	2.52	-	6.73	7.2	-	-	0.22	-	3.85	6.3	-	-
2023	Proposed model	chest	30	-0.32	6.14	6.14	4.86	4.40	0.84	-0.20	5.50	5.50	4.42	5.69	0.80

ME:Mean Error; RMSE:Root Mean Square Error
SD:Standard Deviation; MAE:Mean Absolute Error
MRE:Mean Relative Error; R:Correlation Coefficient

a more comprehensive portrayal of the inherent variability and nonlinearity in the measurements.

D. Comparative Analysis of Model Performance

This section primarily provides a comparative analysis of model performance, specific to non-contact blood pressure measurement methodologies. It is important to note that there are substantial variances across the employed datasets in terms of sensor types and count, test subjects, measurement locations, and even operating frequencies of radars, which include 2.4 GHz, 24 GHz, 300 GHz [41], [42], [44]. The equipment used for data collection encompasses laser radars, dual base station radars, dual split-ring resonators, among other sensing devices [39], [43]. In the case of datasets with relatively few sample subjects, which inherently have a smaller blood pressure range, the performance indicators are likely to yield better results. Therefore, the aim of this comparative analysis is to offer a general performance reference for various non-contact blood pressure measurement methodologies currently in the field.

The detailed comparison of performance metrics is provided in Table V. The units for SBP and DBP are given in mmHg, which is the standard unit of measurement for blood pressure values. Thus, the ME, RMSE, SD, and MAE are also expressed in mmHg, quantifying the magnitude of errors or deviations in the models' blood pressure estimations. The Mean Relative Error (MRE) is presented as a percentage (%), which describes the average relative error with respect to the true values, providing an understanding of the error in terms of proportionate discrepancy. R is a dimensionless quantity that measures the strength and direction of the linear relationship between the estimated and the true blood pressure values. It is a unitless measure as it is derived from the covariance of the variables divided by the product of their standard deviations, resulting in a value that ranges from -1 to 1 . We performed an all-encompassing statistical analysis on the performance results of the model, ME, RMSE, SD, MAE, MRE, and R . The calculation formulas for MRE are given as follows:

$$\text{MRE} = \frac{1}{K} \sum_{k=1}^K \left| \frac{y_k - \hat{y}_k}{y_k} \right| \quad (16)$$

The K is the total number of samples, y_k is the actual value, and $\hat{y}_k$ is the predicted value.

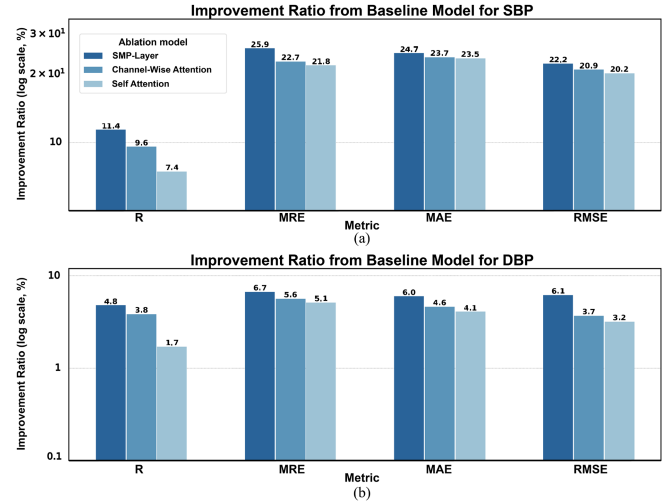


Fig. 12. Logarithmic improvement ratio of ablation study on SMP-Layer, channel-wise attention, and self attention for SBP (a) and DBP (b).

Under the condition of encompassing the maximum number of subjects across various scenarios of motion patterns in the dataset, the proposed model demonstrates better statistical results in terms of MAE, MRE, and correlation coefficients compared to other methods.

V. DISCUSSION

To assess the RSD-Net architecture, experiments were conducted by individually removing the SMP-Layer, channel-wise attention module, and self attention mechanism. The aim of these experiments was to verify the effectiveness of different substructures in the RSD-Net model for blood pressure prediction tasks. In particular, we compared the aforementioned ablation models with the baseline model and presented their performances on various evaluation metrics in the form of graphs. As shown in Fig. 12.

Our adopted evaluation metrics include correlation coefficient, MRE, MAE, and RMSE. Notably, for these metrics, their improvement ratio is calculated differently. For MRE, MAE, and RMSE, which are essentially error metrics where smaller values indicate better model performance, we directly used their relative ratio changes as the improvement ratio. However, for the

Pearson correlation coefficient, as it is a correlation metric where larger values (closer to 1) indicate stronger linear relationships between model predictions and actual values and thus better model performance, we selected the absolute change as the improvement ratio. Finally, as some ablation models may have much higher improvement ratios than others on different metrics, we chose a logarithmic scale to prevent these large values from compressing other indicators in the graph. For the vertical axis of the graph, we used a logarithmic scale to represent the improvement ratio and created separate graphs for SBP and DBP.

A. Assessment of SMP-Layer: Multi-Scale Feature Extraction Capabilities

The design concept of the SMP-Layer is to perform multi-scale feature extraction and fusion on various physiological biomedical signals through a stacked structure of multi-path deformable convolutions. The improvement of the SMP-Layer to the baseline model is shown as the deep blue section in Fig. 12. Due to the unique deformable convolution stack structure of SMP-Layer, compared with channel attention and self attention, achieves the best improvement ratio across all statistical metrics. This corroborates its efficacy in enhancing model performance.

B. Channel-Wise Attention: Impact on Feature Importance Discrimination

The SE block refines feature maps by emphasizing salient features, thereby enhancing the representational capability of the model. Employing global average pooling, the SE block assesses feature map relevance globally, while a fully connected layer with ReLU and Sigmoid activations enables the model to prioritize informative features for blood pressure prediction and diminish reliance on less critical ones. The channel-wise enhancement attributed to this module, as depicted in the blue section of Fig. 12, demonstrates a marked improvement over self attention mechanisms, validating the premise that comprehensive hemodynamic information from physiological signals is more instrumental to model performance than individual signal inputs.

C. Multi-Head Self Attention: Facilitating Information Integration for Optimized Representation

Considering the rich information each multi-path input signal carries, their respective importance may vary, necessitating appropriate weighting and integration for optimal prediction outcomes. The multi-head attention can help the model capture characteristics of the input data and allow the model to automatically adjust its focus based on the importance of these characteristics. The enhancement effect of the this module on the overall model is illustrated as the light blue section in Fig. 12. Among the compared ablation models, its improvement effect is the least notable. This could be attributed to the inadequate extraction of long dependency characteristics of the input signals or the overall contribution of a single signal input being weaker than that of the composite signal pathway characteristics to the model performance enhancement.

VI. CONCLUSION

In this study, we put forth a pioneering approach for pre-processing and segregating composite physiological signals collected via millimeter-wave radar and the RSD-Net used to estimate blood pressure. The model exhibits satisfactory performance in the comparative analysis of statistical parameters across the dataset while concurrently providing an estimate of the general trends in blood pressure fluctuations on an individual sample basis. Performance evaluations were carried out under the BHS international blood pressure measurement standards and got the 'A' grade. However, as all the participants were healthy subjects, the model's performance under hypertensive subjects could not be assessed, suggesting potential early-stage hypertension in only a handful of the participants. Further, our study identified a consistent bias in estimations, likely due to the influence of MSE loss on model outputs under variable input signals. Future work could mitigate this conservative output tendency by refining or augmenting the MSE loss function to better handle input instability.

In essence, our work proposes a new perspective for non-contact blood pressure estimation using composite physiological signals collected via millimeter-wave radar. Supported by embedded hardware platforms, the proposed methods and model can deliver stable blood pressure monitoring that aligns with international blood pressure measurement requirements.

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